METABOLIC & FAT LOSS

Semaglutide — the full review

GLP-1 RECEPTOR AGONIST

What it is

Semaglutide is a lab-made copy of a natural gut hormone called GLP-1, which the body releases after eating. It has been chemically modified — including an amino acid substitution and a fatty-acid side chain that binds to albumin in the blood — so that it remains active for about a week instead of being broken down within minutes. It lowers blood sugar, slows how quickly the stomach empties, and signals appetite centers in the brain.

It is a fully FDA-approved prescription medicine, marketed under several brand names, with an initial US approval in 2017. It has since been evaluated in randomized trials involving tens of thousands of people, including several trials of thousands of participants followed for three to four years — a scale of evidence well beyond most compounds discussed on this site.

Semaglutide is also available as a separate compounded preparation. The shortage that once justified widespread compounding was declared resolved in 2025, and the compounding enforcement-discretion window has since closed; what a patient actually receives — the FDA-approved product or a compounded version — depends on which pathway a clinic or pharmacy is offering.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Type 2 diabetes	**FDA-approved**	One of the approved uses, studied in large placebo-controlled trials; a meaningful reduction in HbA1c is typically seen alongside weight loss.
Long-term weight management	**FDA-approved**	Approved for adults with obesity, or overweight plus a weight-related condition, and separately for adolescents 12 and older with obesity; the effect is smaller in people who also have type 2 diabetes.
Reducing cardiovascular events	**FDA-approved**	Shown to lower the combined rate of heart attack, stroke, and cardiovascular death in a large, multi-year outcome trial in adults with cardiovascular disease and obesity or overweight.
Slowing chronic kidney disease progression in adults with type 2 diabetes	**FDA-approved**	A dedicated outcome trial found a reduced rate of major kidney-disease progression; this indication does not extend to patients without type 2 diabetes.
Knee osteoarthritis pain in people with obesity	**Human trials**	One randomized trial found reduced knee pain alongside weight loss; it remains unclear how much of the benefit is a specific drug effect versus a general consequence of losing weight off a weight-bearing joint. Not an approved indication.

USED FOR	EVIDENCE	WHAT THAT MEANS
Reducing alcohol craving and heavy drinking	**Limited human data**	A small randomized trial in people not seeking treatment for alcohol use found reduced craving and fewer heavy-drinking days over nine weeks at the lowest clinical dose. The investigators described the finding as preliminary. Not an approved indication.
Slowing Alzheimer's disease or cognitive decline	**Human trials**	Tested at scale in more than 3,800 people with mild cognitive impairment or early Alzheimer's disease; the trials improved certain disease-related biomarkers but did not meet their primary endpoint of delaying disease progression.

How it works

The intestine naturally releases GLP-1 whenever food arrives. GLP-1 does three things at once: it tells the pancreas to release insulin, but only when blood sugar is actually high; it tells the pancreas to stop releasing the hormone that pushes blood sugar up; and it tells the stomach to empty more slowly and the brain that the body is full.

Natural GLP-1 breaks down within minutes. Semaglutide is that same signal rebuilt so the body cannot clear it quickly — one weekly injection (or a daily tablet, co-formulated with an absorption enhancer) keeps the fullness and blood-sugar signal active for roughly a week at a time.

The effect lasts only as long as the medicine is taken — it functions as an ongoing treatment for a chronic condition rather than a one-time reset. Most common side effects, such as nausea and constipation, are the same mechanism turned up too high, which is why dosing is increased gradually over months.

What the research actually looked at

Semaglutide has one of the largest clinical trial evidence bases of any compound discussed on this site: FDA approval across several indications rests on separate randomized, controlled trial programs that together randomized tens of thousands of people, including multi-year outcome trials that measured hard endpoints such as cardiovascular events and kidney disease progression, not just weight or blood sugar.

Effectiveness varies meaningfully by population. Weight-loss results reported in trials of people with type 2 diabetes have been smaller than in trials of people without diabetes, and some approvals — including a 2025 approval for a form of fatty liver disease — rest on an accelerated pathway using a biomarker endpoint, with confirmatory long-term outcome data still pending.

None of this trial evidence transfers to compounded semaglutide, which by definition is not the specific product that was studied. Some other investigated uses have far thinner evidence, ranging from a single small trial to no dedicated trial at all, and a large randomized trial specifically testing semaglutide for slowing Alzheimer's disease did not meet its primary endpoint, despite improving certain disease-related biomarkers.

Common misunderstandings

- **“Compounded semaglutide is the same drug as the approved product, just cheaper.”** FDA has stated this claim is false and has taken enforcement action against sellers making it. Compounded versions are made by a pharmacy from bulk material and

have not been reviewed by FDA, and some formulations use different salt forms than the approved drug.

- **“If it worked in the trials, that proves the compounded version works too.”** None of the tens of thousands of people randomized in semaglutide trials received a compounded product — every efficacy figure belongs to the FDA-approved product made under approved manufacturing controls.
- **“Microdosing gives the same benefits with fewer side effects.”** There is no trial of sub-therapeutic semaglutide dosing in metabolically healthy people; every efficacy result comes from full-titration dosing in people who met specific trial entry criteria.
- **“It's approved to prevent or slow Alzheimer's disease.”** A large randomized trial tested semaglutide for this specific purpose and did not meet its primary endpoint, despite improving certain disease-related biomarkers.

Who it is not for

- Anyone with a personal or family history of medullary thyroid cancer, or MEN 2 — an absolute contraindication.
- Anyone with a serious allergic reaction to semaglutide or its ingredients.
- Pregnant or breastfeeding patients, or anyone trying to conceive — the long half-life means it should be stopped well before a planned pregnancy.
- Anyone with a history of pancreatitis or gallbladder disease, without discussing it first.
- Anyone scheduled for a procedure requiring general anesthesia or sedation, without informing the anesthesia team in advance.
- Anyone sourcing product from an unverified online seller rather than a licensed pharmacy.

Competitive athletes: GLP-1 medicines including semaglutide are not currently prohibited in sport, and a therapeutic use exemption is not required to use an FDA-approved product as prescribed. Semaglutide has, however, been placed on WADA's Monitoring Program, meaning testing laboratories track its use even though it is not a violation — and the more practical risk for athletes lies in sourcing an unapproved or compounded product of uncertain content rather than in the approved medicine itself.

Approved product, or a compounded preparation — and the difference matters

This is an FDA-approved medicine. It is available as the manufacturer's own approved product, and — separately — as a compounded preparation. Which one is being offered is always stated plainly.

If the approved product is right, that is what gets prescribed:

- **The manufacturer's pen or vial.** Reviewed by the FDA for safety, effectiveness and quality, exactly as labeled.
- **Filled at a licensed pharmacy.** Dispensed from your prescription like any other approved medicine.

If a compounded preparation is clinically appropriate instead, it is prepared by a licensed 503A compounding pharmacy:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not itself an FDA-approved product — but it is still made under a licensed pharmacy's enforceable standards, not an unregulated website.

Where this stands

A fully FDA-approved medicine with an unusually large evidence base. Approved across several indications including type 2 diabetes, weight management, and cardiovascular risk reduction, semaglutide has been studied in randomized trials involving tens of thousands of people, including multi-year outcome trials — a scale of evidence well beyond most compounds in this category.

Compounded semaglutide is a separate, unapproved product. The compounding shortage was declared resolved in 2025 and the enforcement-discretion window has since closed. FDA has issued warning letters to sellers claiming a compounded version is the same as the approved drug.

Ask whether you are being offered the approved product or a compounded preparation, and if compounded, why.

Clinical detail

Approved under several brand names, with an initial US approval in 2017; indications now include type 2 diabetes, chronic weight management, cardiovascular risk reduction, chronic kidney disease progression, and, under an accelerated approval, a form of fatty liver disease.

Structurally modified from natural GLP-1 with an amino acid substitution and a fatty-acid side chain that binds albumin, extending its activity to roughly a week and enabling once-weekly injection; a once-daily oral formulation, co-formulated with an absorption enhancer, also exists.

Development and post-approval trials have randomized tens of thousands of people, including outcome trials following participants for three to four years.

The compounding shortage was declared resolved in 2025, after which the legal basis for routine compounding narrowed considerably.

A large randomized trial testing semaglutide specifically for slowing Alzheimer's disease did not meet its primary endpoint despite improvements on certain disease biomarkers — a notable example of biomarker change not translating into clinical benefit.

> The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.
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> Semaglutide is FDA-approved for the uses described as approved above. Off-label uses discussed here are not FDA-approved uses. A compounded preparation, where used instead of the approved product, is not itself an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.
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> Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any unapproved purpose.
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> Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.